

Recitation 2

Causal Inference, Confounding, and Simpson's Paradox
Review and Practice Problems

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1 Potential outcomes and causal effects

Let the binary treatment $\tilde{t}$ equal 1 for treatment and 0 for control. Every unit has two **potential outcomes**:

$$\widetilde{po}_0 = \text{outcome under control}, \quad \widetilde{po}_1 = \text{outcome under treatment}.$$

Both are defined whether or not the unit actually receives treatment. The observed outcome is

$$\tilde{y} = \begin{cases} \widetilde{po}_0, & \tilde{t} = 0, \\ \widetilde{po}_1, & \tilde{t} = 1, \end{cases} \quad \text{or equivalently} \quad \tilde{y} = (1 - \tilde{t})\widetilde{po}_0 + \tilde{t}\widetilde{po}_1.$$

This relationship is called **consistency**. For one unit we observe only one potential outcome; the other is a **counterfactual**. This is the fundamental problem of causal inference.

For a binary outcome, the population causal risk difference is

$$\Delta_{\text{causal}} := \mathbb{P}(\widetilde{po}_1 = 1) - \mathbb{P}(\widetilde{po}_0 = 1) = \mathbb{E}[\widetilde{po}_1] - \mathbb{E}[\widetilde{po}_0].$$

The difference visible in observational data is instead

$$\Delta_{\text{obs}} := \mathbb{P}(\tilde{y} = 1 \mid \tilde{t} = 1) - \mathbb{P}(\tilde{y} = 1 \mid \tilde{t} = 0).$$

By consistency, the first term describes $\widetilde{po}_1$ among treated units and the second describes $\widetilde{po}_0$ among control units. These are generally different subpopulations, so association need not equal causation.

If treatment is independent of both potential outcomes,

$$\tilde{t} \perp\!\!\!\perp \{\widetilde{po}_0, \widetilde{po}_1\},$$

then $\Delta_{\text{obs}} = \Delta_{\text{causal}}$ at the population level. Random assignment creates this independence by design. In a finite randomized sample, the two observed rates are still estimates and need not equal the population probabilities exactly.

2 Confounding and adjustment

A pre-treatment variable $\tilde{a}$ is a **confounder** when it helps create dependence between treatment and the potential outcomes. For example, age may affect which patients take a drug and may also predict recovery under either treatment. Comparing treated and control patients without accounting for age then mixes together treatment differences and age-composition differences.

Suppose the following assumptions are plausible:

1. **Consistency:** the observed outcome equals the potential outcome under the treatment received.
2. **Conditional independence:** $\{\widetilde{\text{po}}_0, \widetilde{\text{po}}_1\} \perp\!\!\!\perp \widetilde{t} \mid \widetilde{a}$.
3. **Positivity:** within every relevant value of a , both treatment groups can occur.

Then, for $r \in \{0, 1\}$,

$$\mathbb{P}(\widetilde{\text{po}}_r = 1) = \sum_a \mathbb{P}(\widetilde{a} = a) \mathbb{P}(\widetilde{y} = 1 \mid \widetilde{a} = a, \widetilde{t} = r).$$

The same population weights $\mathbb{P}(\widetilde{a} = a)$ are used for both treatments. By contrast, the unadjusted observed rate is

$$\mathbb{P}(\widetilde{y} = 1 \mid \widetilde{t} = r) = \sum_a \mathbb{P}(\widetilde{a} = a \mid \widetilde{t} = r) \mathbb{P}(\widetilde{y} = 1 \mid \widetilde{a} = a, \widetilde{t} = r),$$

which may use very different weights in the two treatment groups. This is the law of total probability, and the changing weights are the mechanism behind Simpson’s paradox.

3 Simpson’s paradox: Curry versus Lee

During the 2014–2015 NBA season, Courtney Lee had the higher overall three-point percentage, but Stephen Curry had the higher percentage within both distance categories:

Shot type	Stephen Curry	Courtney Lee
Short threes (≤ 24 feet)	45/90 = 50.0%	56/116 = 48.3%
Long threes (> 24 feet)	145/366 = 39.6%	19/55 = 34.5%
All threes	190/456 = 41.7%	75/171 = 43.9%

This reversal is **Simpson’s paradox**. Shot distance, denoted $\widetilde{d}$, explains it. Lee took 116/171 = 67.8% of his attempts from the easier short range, while Curry took only 90/456 = 19.7% from that range. Each overall percentage is a weighted average with a different set of weights. For example,

$$\mathbb{P}(\widetilde{y} = 1 \mid \widetilde{t} = \text{Lee}) = 0.678(0.483) + 0.322(0.345) = 0.439.$$

For a common comparison, weight both players by the shot mix in the combined data: 206/627 = 0.329 short and 421/627 = 0.671 long. The standardized rates are

$$\text{Curry: } 0.329(0.500) + 0.671(0.396) = 0.430,$$

$$\text{Lee: } 0.329(0.483) + 0.671(0.345) = 0.390.$$

Standardization therefore favors Curry. It is a useful comparison of shooting under a common distance distribution. Interpreting it as a causal comparison requires the strong assumption that, conditional on distance, player identity is independent of the relevant potential outcomes; other shot characteristics may violate that assumption.

Read the comparison before interpreting it. Marginal rates answer “what happened under each group’s actual mix?” Stratum-specific rates answer “what happened within a fixed category?” Standardized rates answer “what would the rates be under a common mix?”

Practice problems

Problem 1: What is observed?

[7 minutes]

Four patients have binary treatment and recovery outcomes:

Patient	Treatment t	Observed y	po_0	po_1
A	0	0		
B	0	1		
C	1	1		
D	1	0		

- (a) Fill in every potential outcome determined by the observed data. Mark each unobserved potential outcome with “?”.
- (b) Compute the observed difference in recovery rates, treated minus control.
- (c) Assuming only that potential outcomes are binary, find the smallest and largest possible finite-sample average causal effect $\frac{1}{4} \sum_{i=1}^4 (po_{1,i} - po_{0,i})$.
- (d) If treatment was randomly assigned, does that reveal each missing counterfactual? What does randomization provide instead?

Problem 2: Diagnose the Curry–Lee reversal**[8 minutes]**

Use the basketball table in the review.

- (a) Which player has the higher marginal three-point percentage? Which player has the higher percentage within each distance category?
- (b) Write Curry’s overall percentage as a weighted average of his short- and long-shot percentages. Compare his weights with Lee’s.
- (c) In one or two sentences, explain why the direction reverses after the data are stratified by distance.
- (d) Using a target shot mix of 40% short and 60% long, compute a standardized percentage for each player. Which is larger?
- (e) Does part (d) by itself prove that Curry would outperform Lee on the same shots? State the additional assumption needed for that causal claim.

Problem 3: A drug and patient risk**[8 minutes]**

An observational study records the following recoveries:

Patient group	Drug ($t = 1$)	No drug ($t = 0$)
Lower risk	$45/50 = 90\%$	$80/100 = 80\%$
Higher risk	$30/100 = 30\%$	$10/50 = 20\%$
All patients	$75/150 = 50\%$	$90/150 = 60\%$

- (a) Compute the unadjusted observed risk difference. What conclusion would it suggest?
- (b) Compute the risk difference within each risk group. Describe the reversal.
- (c) In the full sample, half the patients are lower risk and half are higher risk. Standardize both treatment groups to this common 50/50 mix. Compute the adjusted risk difference.
- (d) Explain how risk group creates confounding here. State the assumptions needed to interpret the adjusted difference causally.

Problem 4 (backup): True or false?

Optional if time allows. Suggested working time: 5–7 minutes.

- (a) Random assignment reveals both potential outcomes for every unit.
- (b) If treatment has a positive risk difference in every stratum, then any standardized risk difference using the same nonnegative stratum weights for both treatments is positive.
- (c) To adjust for a confounder, use the treated group's stratum weights for the treated rate and the control group's weights for the control rate.
- (d) Seeing a Simpson reversal proves that one of the two marginal groups has been measured incorrectly.

Takeaways

- Potential outcomes define causal questions, but only one is observed per unit.
- Randomization breaks dependence between treatment and potential outcomes.
- Confounding makes observed treatment groups incomparable; conditional comparisons and common-weight standardization can repair this only under substantive assumptions.
- Simpson's paradox is a weighted-average reversal, not an arithmetic error.

Sources

The review follows the course handouts *Causal Inference* and *Modeling Multiple Discrete Variables*, together with [Was Courtney Lee Better Than Stephen Curry? Simpson's Paradox](#) from *Probability and Statistics for Data Science*.